

PRODUCT REQUIREMENTS DOCUMENT

PartsDex

Plumbing, HVAC & Boiler Parts Identifier

MVP Build — Capstone Project

Version 1.0 | March 2026

Status	DRAFT — ACTIVE
Timeline	2 Weeks (MVP)
Team Size	2 Developers

1. Executive Summary

PartsDex is a mobile-first Progressive Web App (PWA) designed for on-site crews and trade professionals in the plumbing, HVAC, and boiler & heating industries. Built primarily for field use on smartphones, PartsDex also works seamlessly on desktop browsers — install it once from the browser, and it lives on your phone like a native app. The application solves a persistent real-world problem: in these trades, the same physical part can have dozens of different names, part numbers, and regional aliases — making identification, ordering, and communication extremely difficult for both experienced technicians and new entrants to the field.

PartsDex provides a clean, searchable database of parts with detailed information including industry names and aliases, technical diagrams, compatibility guides, and use-case overviews. The MVP is scoped for a 2-week build by a 2-person frontend-focused team, with a clear path to a future AR integration phase.

1.1 Problem Statement

Tradespeople and apprentices in plumbing, HVAC, and boiler & heating regularly encounter the following friction points:

- A single part (e.g., a compression fitting) may be called by 6+ different names depending on region, manufacturer, and trade era
- New technicians waste time and money ordering wrong parts due to naming confusion
- There is no single, visual, searchable reference that bridges old and new terminology
- Existing resources (manufacturer PDFs, trade catalogs) are fragmented, offline, and not user-friendly

1.2 Solution

PartsDex centralizes part knowledge in a fast, visual, PWA built for the job site. A technician pulls out their phone, searches or browses, and instantly has a complete part profile in hand — what it is, what it does, what it's compatible with, how it's installed, and every name it goes by in the industry. No app store download required; it installs directly from the browser and works offline.

1.3 Target Users

User Type	Background	Primary Need
Field Technician	5–20+ years in the trade	Fast part ID and alias lookup on-site
Apprentice / New Hire	0–2 years experience	Learning part names, uses, and compatibility
Parts Counter Staff	Supply house / distributor	Cross-referencing customer descriptions to SKUs

User Type	Background	Primary Need
HVAC/Plumbing Instructor	Trade school educator	Visual teaching aids and reference material

2. Product Overview

2.1 Product Vision

To be the definitive on-the-job reference tool for trade professionals — the app you open when you're holding a part and don't know what it's called, where it goes, or what it connects to.

2.2 MVP Scope

The MVP focuses on delivering a complete, polished core experience across three industry verticals. Feature scope has been deliberately constrained to ensure a shippable, high-quality product within 2 weeks.

In Scope for MVP

- Part search (by name, alias, part number, keyword)
- Part detail pages (overview, diagram, aliases, compatibility, use cases)
- Category browsing by industry vertical (Plumbing, HVAC, Boiler & Heating)
- Progressive Web App (PWA) — installable on iPhone and Android, works offline, no App Store required
- Static data layer with curated initial parts library
- Filter and sort by category, type, and material

Out of Scope for MVP (Future Phases)

- AR part identification via camera
- User accounts and saved parts
- Community-submitted parts or edits
- Native iOS / Android app (PWA covers this need for MVP; native app is a future phase if needed)
- API integrations with supplier catalogs (e.g., Ferguson, Grainger)
- AI-powered part identification from photo input

2.3 Success Metrics for MVP

Metric	MVP Target
Parts in database at launch	100+ curated parts across 3 verticals
Search result accuracy	Correct part returned in top 3 results for all seeded queries
Mobile usability	All core flows usable on a 375px wide viewport
Page load time	< 2 seconds on 4G connection

Metric	MVP Target
Capstone evaluation	Demonstrates real-world usefulness to non-technical evaluators

3. Feature Specifications

3.1 Feature 1 — Global Search

The search bar is the primary entry point for the app and must support flexible, forgiving queries.

Functional Requirements

- Search input is visible on every page (persistent header)
- Searches run across: part name, all aliases, part number, category, and description text
- Results display within 300ms for the seeded local dataset
- Results show part thumbnail, primary name, and category badge
- 'Did you mean?' suggestion for near-matches and typos
- Empty state shows recently viewed parts or popular searches

Search Behavior

- Partial match: searching 'comp' should surface 'compression fitting', 'compression valve'
- Alias match: searching 'olive' should return 'compression ring' (olive is a common alias)
- Case-insensitive, accent-insensitive
- Minimum 2 characters to trigger search

3.2 Feature 2 — Part Detail Page

The part detail page is the core product experience. Every part has a dedicated page structured into clear, scannable sections.

Page Sections

Section	Content	Priority
Part Header	Primary name, category badge, short tagline description	P0 — Must Have
Diagram	Technical illustration or photo showing the part clearly labeled	P0 — Must Have
Overview	What the part is, what it does, typical applications	P0 — Must Have
Industry Names & Aliases	All known names: trade slang, regional terms, manufacturer names, old vs. new terminology	P0 — Must Have
Compatible Parts	Parts this component connects to or is used with	P1 — Should Have
Installation Notes	Brief overview of typical installation context	P1 — Should Have

Section	Content	Priority
Available Materials	Copper, PVC, brass, stainless, etc.	P1 — Should Have
Related Parts	Similar or commonly confused parts with link to their detail pages	P2 — Nice to Have

3.3 Feature 3 — Category Browser

Users who don't know the part name can browse by industry vertical and part category.

Industry Verticals

- Plumbing — pipes, fittings, valves, fixtures, drainage, supply
- HVAC — ductwork, air handlers, refrigerant components, thermostats, filters, dampers
- Boiler & Heating — boiler components, expansion tanks, circulators, radiators, controls, heat exchangers

Browse UX

- Landing page displays 3 vertical cards with icons and part counts
- Selecting a vertical shows part type sub-categories (e.g., under Plumbing: Fittings, Valves, Pipes, Fixtures)
- Part grid view shows thumbnail, name, and short description
- Filter panel allows narrowing by material, size range, and connection type

3.4 Feature 4 — Alias & Terminology Glossary

A standalone glossary section for users who want to explore terminology without a specific part in mind.

- Alphabetical list of all known aliases and trade terms in the database
- Each alias entry links to its canonical part detail page
- Helpful for new technicians learning trade vocabulary
- Filterable by industry vertical

3.5 Feature 5 — PWA & Mobile-First Experience

PartsDex is built as a Progressive Web App (PWA) with a mobile-first design philosophy. The primary use case is a field technician on a job site pulling out their phone to identify an unfamiliar part or look up what they need to complete a job. The app installs directly from Chrome or Safari onto any smartphone home screen — no App Store, no friction.

- Designed and tested at 375px viewport width (iPhone SE — the smallest common smartphone) as the primary design target; scales up for tablet and desktop
- Diagrams scale and remain legible on small screens
- Touch targets minimum 44x44px per WCAG guidelines
- Bottom navigation bar for one-handed, thumb-friendly access on phones
- Search bar stays pinned to top on scroll
- PWA installable directly from Chrome or Safari — appears on phone home screen like a native app, no App Store required
- Offline mode via service worker — critical parts data cached locally so crews can use the app in basements, mechanical rooms, and areas with no cell signal
- Works on both iPhone (Safari) and Android (Chrome) — same codebase, same experience
- Desktop web also fully supported for office staff, instructors, and parts counter use

4. Technical Architecture & Stack

4.1 Recommended Tech Stack

The stack is chosen to maximize velocity for a frontend-focused team while delivering a polished, performant result.

Layer	Technology	Rationale
UI Framework	React (Vite) + PWA Plugin	Fast dev server, component reuse, both teammates likely familiar
PWA Layer	vite-plugin-pwa + Workbox	Generates service worker and web manifest; enables offline mode, install-to-home-screen on iOS and Android
Styling	Tailwind CSS	Rapid, consistent styling without writing custom CSS files
Routing	React Router v6	Client-side routing for part detail pages and category pages
Data Layer	Local JSON / static files	No backend needed for MVP; fast to seed and update
Search Engine	Fuse.js	Lightweight fuzzy search library — works entirely in-browser on JSON
Icons	Lucide React	Clean, consistent icon set with plumbing/HVAC adjacent icons
Deployment	Vercel	Free tier, instant deploys from GitHub, zero config for Vite/PWA apps; serves the installable PWA manifest
Version Control	GitHub	Collaboration between 2 developers

4.2 Data Architecture

Each part is stored as a structured JSON object in a flat file. This eliminates backend complexity while keeping the data easy to edit, extend, and eventually migrate to a real database.

Part Data Schema

Field	Type	Description
id	string	Unique slug identifier (e.g., 'compression-fitting-15mm')
primaryName	string	The canonical, most widely recognized name
aliases	string[]	All known trade names, regional names, slang, and manufacturer terms

Field	Type	Description
category	string	Top-level: 'plumbing' 'hvac' 'boiler-heating'
subCategory	string	e.g., 'fittings', 'valves', 'ductwork', 'controls'
overview	string	Plain-English description of what the part is and does
diagramUrl	string	Path to diagram SVG or PNG image asset
compatibleWith	string[]	Array of part IDs that this part connects with
materials	string[]	e.g., ['copper', 'brass', 'PVC', 'stainless']
installationNotes	string	Brief installation context
tags	string[]	Free-form tags for improved search (e.g., 'compression', 'push-fit', 'solder')

4.3 Project Structure

Recommended folder structure for the React/Vite project:

```
src/ components/ - reusable UI components (SearchBar, PartCard,
DiagramViewer)
  pages/ - route-level pages (Home, PartDetail, Browse, Glossary)
  data/ - parts.json, categories.json, aliases.json
  assets/ - diagrams, icons, images
  hooks/ - useSearch, usePartById, useCategory
  utils/ - search config (Fuse.js), data helpers
```

5. 2-Week Build Plan

5.1 Sprint Overview

The 2-week sprint is divided into two focused phases: build week and polish week. Each day has defined deliverables to prevent scope drift and ensure a shippable product at the end of Week 2.

5.2 Week 1 — Core Build (Days 1–7)

Day	Focus	Deliverables	Owner Suggestion
Day 1	Project Setup	Repo, Vite + React + Tailwind + vite-plugin-pwa, routing scaffold, PWA manifest & service worker configured, deploy pipeline to Vercel	Both
Day 2	Data Layer	parts.json schema finalized, 25 parts seeded across all 3 verticals, Fuse.js integrated	Dev 1
Day 2	UI Foundation	Color system, typography scale, navigation bar, layout shell	Dev 2
Day 3	Search Feature	Search bar component, results page, fuzzy match working, alias search confirmed	Dev 1
Day 3	Home Page	Hero section, 3 vertical cards, search CTA, recent/popular parts	Dev 2
Day 4	Part Detail Page	All P0 sections live: header, diagram, overview, aliases table	Dev 2
Day 4	Data Expansion	Expand to 60 parts total; add diagrams/images for first 25	Dev 1
Day 5	Category Browser	Vertical landing pages, sub-category grid, part cards	Dev 2
Day 5	Data Expansion	Complete 100 parts; all with aliases and compatibility data	Dev 1
Day 6	Compatible Parts	Compatible parts section on detail page; related parts links	Dev 2
Day 6	Filter System	Category filter, material filter, vertical filter on browse page	Dev 1
Day 7	Glossary Page	Alias glossary A–Z, filter by vertical, links to part pages	Dev 1
Day 7	Mobile Pass	First full mobile pass; bottom nav, touch targets, test	Dev 2

Day	Focus	Deliverables	Owner Suggestion
		install-to-home-screen on iOS and Android	

5.3 Week 2 — Polish & Finalize (Days 8–14)

Day	Focus	Deliverables
Day 8	UI Polish	Consistent spacing, color, typography across all pages; diagram styling
Day 9	PWA & Mobile Finalize	Full mobile audit on all pages; fix overflow issues, verify offline mode works, test on real iOS and Android devices
Day 10	Empty States & Errors	No-results state, loading states, graceful fallbacks for missing images
Day 11	Content Audit	Review all 100 parts for accuracy, completeness, and clear writing
Day 12	Performance	Optimize images, lazy load diagrams, verify <2s load time
Day 13	QA & Bug Fix	Full walkthrough of all user flows; fix any remaining bugs
Day 14	Demo Prep	Final deploy, demo script prepared, capstone presentation materials

5.4 Risk Register

Risk	Severity	Mitigation
Data sourcing: finding 100 accurate, well-described parts	High	Start data entry Day 1; use trade catalogs (e.g., Wolseley, Ferguson) as reference; prioritize depth over breadth
Diagram/image quality: finding clear technical diagrams	Medium	Use royalty-free SVGs; create simple labeled diagrams in Figma or Excalidraw if needed
Scope creep: adding features mid-sprint	Medium	Strictly follow this PRD; log new ideas in a backlog doc, do not build during MVP sprint
Search accuracy on aliases	Low	Fuse.js threshold tuning; test with 20 real-world alias queries before Day 5

6. UX & Design Guidelines

6.1 Design Principles

- Clarity over cleverness — field technicians need information fast, not beautiful animations
- Mobile-first — design for a phone screen in a dirty hand, not a desktop browser
- Scannable — use clear labels, visual hierarchy, and whitespace so users find info without reading
- Trust through accuracy — every part entry must be factually correct; wrong info destroys credibility

6.2 Color System

Role	Hex Value	Usage
Primary Blue	#1B4F8A	Navigation, headings, primary CTAs, category badges
Success Green	#2E7D32	Compatibility indicators, active states, status labels
Background Gray	#F5F5F5	Page background, card backgrounds, alternate rows
Text Dark	#1A1A2E	Body text, part names, primary labels
Text Muted	#555555	Secondary text, metadata, timestamps

6.3 Key UI Components

Part Card (used in grid/search results)

- Part thumbnail diagram (square, 1:1 ratio)
- Primary name (bold, 16px)
- Category badge (color-coded pill)
- One-line description excerpt
- Tap/click navigates to Part Detail page

Search Bar

- Full-width, persistent in header
- Placeholder: 'Search by name, alias, or part number...'
- Results dropdown appears below as user types (debounced 200ms)
- Shows part thumbnail + name + category in dropdown

Alias Table (on Part Detail page)

- Two columns: 'Name / Alias' and 'Context'
- Context explains where that name is used (e.g., 'UK trade term', 'Old BS standard', 'Manufacturer brand name')
- Highlighted row for the primary/canonical name

7. Future Phases (Post-MVP)

7.1 Phase 2 — AR Part Identification

The most compelling differentiator for PartsDex's long-term roadmap. A technician points their phone camera at an unknown part, and the app identifies it.

- Implement camera access via browser MediaDevices API (already supported in PWA context on iOS and Android)
- Build or integrate a parts image classification model (TensorFlow.js for in-browser, or API call to a hosted model)
- Initial training data: photos of the 100 MVP parts from multiple angles
- AR overlay displays part name, aliases, and compatibility hints directly on camera feed
- Estimated complexity: 4–6 weeks additional development

7.2 Phase 3 — Backend & User Accounts

- Migrate from static JSON to a hosted database (Supabase or Firebase recommended)
- User accounts with saved/favorited parts
- Technician-submitted aliases and notes (community knowledge layer)
- Admin dashboard for content management

7.3 Phase 4 — Supplier Integration

- Real-time inventory and pricing from major distributors (Ferguson, Grainger, Wolseley)
- 'Where to buy this part' links with local availability
- Part number cross-reference between manufacturer catalogs

7.4 Phase 5 — Native Mobile App

- iOS and Android native apps via React Native if PWA experience needs improvement on specific platforms (large logic reuse from PWA MVP)
- Offline mode for use in basements, mechanical rooms, and areas without cell signal
- Push notifications for new parts in a technician's saved categories

8. Appendix

8.1 Sample Parts List for MVP Seeding

The following is a starter list of parts to seed across all three verticals. Each should have a full entry including aliases, overview, and compatibility.

Plumbing (suggested: 40 parts)

- Compression fitting (olive, compression ring, compression nut)
- Push-fit fitting (Speedfit, SharkBite, push-to-connect)
- Gate valve (sluice valve, full-bore valve)
- Ball valve (quarter-turn valve, shut-off valve)
- Stop valve (service valve, mini valve)
- P-trap (waste trap, sink trap, bottle trap)
- Pipe coupling, elbow (street elbow, sweep elbow), tee, reducer
- Float valve (ball cock, ballvalve, toilet fill valve)
- Isolation valve, check valve (one-way valve, non-return valve)
- Flexible tap connector, basin wrench, pipe clip/bracket

HVAC (suggested: 35 parts)

- Schrader valve (service valve, charging valve, AC valve)
- TXV — thermostatic expansion valve (metering device, expansion valve)
- Filter drier (liquid line drier, moisture indicator)
- Air handler unit (fan coil unit, AHU, air handling unit)
- Damper (volume control damper, fire damper, VAV box)
- Duct elbow, duct reducer, plenum, diffuser, register
- Condenser fan motor, evaporator coil, compressor
- Refrigerant sight glass, service port, Schrader core remover

Boiler & Heating (suggested: 25 parts)

- Expansion vessel (expansion tank, pressure vessel, diaphragm tank)
- Circulator pump (heating pump, zone pump, wet rotor pump)
- Pressure relief valve (PRV, safety valve, T&P valve)
- Zone valve (motorized valve, 2-port valve, mid-position valve)
- Heat exchanger (plate heat exchanger, PHE, brazed plate)
- Lockshield valve (balancing valve, radiator lockshield)
- TRV — thermostatic radiator valve (rad valve, room thermostat valve)
- Filling loop (filling link, bypass link, system filler)

8.2 Useful Reference Resources for Data Entry

- CIPHE (Chartered Institute of Plumbing and Heating Engineering) — parts glossaries
- Ferguson Enterprises product catalog — cross-reference part numbers
- Grainger industrial catalog — HVAC and boiler component descriptions
- ASHRAE Handbook — authoritative HVAC terminology
- BS EN standards documentation — European/UK part naming conventions

8.3 Document Revision History

Version	Date	Author	Changes
1.0	March 2026	Capstone Team	Initial PRD — full MVP specification